



MCT231: Basics of Mechatronics Team 7

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I. Introduction

This project presents a mechatronics-based interactive game designed for children, combining mechanical systems, control, and basic physics principles. The system uses a ball-based mechanism where balls are launched, lifted, and guided through a structured path. The objective of the game is to engage users in a fun and educational way while introducing them to concepts such as motion, control, and decision-making.

The game incorporates multiple subsystems including a shooting mechanism, a lifting mechanism, and a sorting/scoring section. These elements work together to create a dynamic and interactive environment that enhances both entertainment and learning.

Project Overview & Concept

The project is a smart mechanical game that uses two types of balls:

- White balls → increase the player's score
- Black balls → decrease the player's score

The system begins with a feeding unit that releases balls into a shooting mechanism, which launches the ball along a defined path. The ball is then transferred through a lifting mechanism that elevates it to a higher level, allowing it to continue its motion through the system.

At the final stage, the ball reaches a decision area controlled by a movable slider (cylinder) operated by the player. The player must adjust the slider position to guide the ball into the desired path based on its colour: directing a white ball to the correct slot increases the score, while directing a black ball incorrectly reduces the score. This introduces an element of timing, control, and quick decision-making, making the game both challenging and engaging.

Concept and Objectives

The main concept of the project is to integrate mechanical motion with user interaction to create a responsive system. The objectives include:

- Designing a fully integrated mechatronic system
- Demonstrating projectile motion and ball dynamics

- Developing mechanical mechanisms (shooting, lifting, guiding)
- Enhancing user interaction and reflex-based decision-making
- Creating an educational and entertaining game for children

2. User Research

Research Methodology

Before finalising the toy concept, the team conducted a structured user research phase to validate the design direction. The research aimed to identify what children aged 2–10 and their parents value most in an interactive toy, and to understand the competitive landscape of educational mechatronic toys.

Competitive Analysis

Existing toys included marble-run sets, pinball machines, and simple robotic kits. Key gaps identified:

- Most toys lack real-time scoring feedback.
- Few toys combine a colour-sensing decision mechanic with physical launching.
- Higher-end STEM toys exceed the target price point for most families.

These findings directly shaped the final concept: an affordable, colour-sorting ball game with a live score display, enclosed wiring, and no small detachable parts.

3. Design Challenge

The central design challenge was to create a mechatronic toy that is simultaneously safe for young children, educationally meaningful, mechanically reliable, and manufacturable within a budget of approximately 14,000 L.E.

Key Constraints

- Safety: no exposed electrical contacts, no sharp corners, ball diameter large enough to prevent choking (≥ 30 mm).
- Size: overall footprint ≤ 100 cm $\times$ 100 cm, height ≤ 80 cm.
- Cost: primary structure from 3D-printed PLA and laser-cut plywood.
- Complexity: must be understandable and maintainable by a non-specialist adult.
- Durability: designed to survive repeated use and occasional impacts from young children.

Identified Trade-offs

The team had to balance several competing demands throughout the design process:

- Sorting speed vs. detection accuracy: faster movement of the slider increases excitement but reduces the window for the IR to read the ball correctly.
- Compactness vs. mechanism clearance: a smaller footprint limits the travel length available for the lifting screw and the projectile range of the shooting arm.
- Motor power vs. noise: higher-torque motors are noisier, which may disturb younger children or parents.
- 3D-print resolution vs. print time: finer prints give better fitment but significantly increase fabrication time.

4. Ideation

Ideation Process Overview

The team used a structured diverge-then-converge approach to idea generation. Four main techniques were employed: mind mapping, mood boarding and a weighted decision matrix.

Mind Map

Starting from the central theme 'interactive children's toy', the team branched out into four domains: Mechanisms (shooting, lifting, sorting, guiding), Electronics (sensors, microcontrollers, actuators, displays), Interaction (player input, feedback, scoring), and Safety (materials, enclosures, edge treatment). Each domain was further expanded to individual components and ideas before being pruned by feasibility.

Mood Board

Visual references were collected from existing ball-track toys, pinball machines, factory conveyor systems, and colorful classroom learning tools. The mood board established the design language: rounded forms, bold primary colors on a dark navy/green background, visible mechanical motion, and clear score displays.

Decision Matrix

The final concept selection used a weighted matrix scoring each concept on: Safety (30%), Educational Value (25%), Feasibility (25%), and Engagement (20%). This concept scored highest overall (4.1/5), confirming the team's preferred direction.

5. Toy Description and Functionality

5.1 Mechanical Elements

The toy consists of three primary mechanical subsystems: the lifting mechanism, the shooting mechanism, and the sorting/guiding mechanism. All structural components are 3D-printed in PLA or laser-cut from 6 mm plywood.

Lifting Mechanism – Archimedes Screw

A helical screw (length 400 mm, outer diameter 100 mm, pitch 45 mm) transports balls from the base collection area up to the shooter entry point. The screw is driven by a DC yellow motor (stall torque 0.08–0.15 Nm). Actuator sizing confirmed the required motor torque of 0.03 Nm, well within the motor's capability. T_{RMS} was calculated at 0.0202 Nm for a 7-second lift cycle.

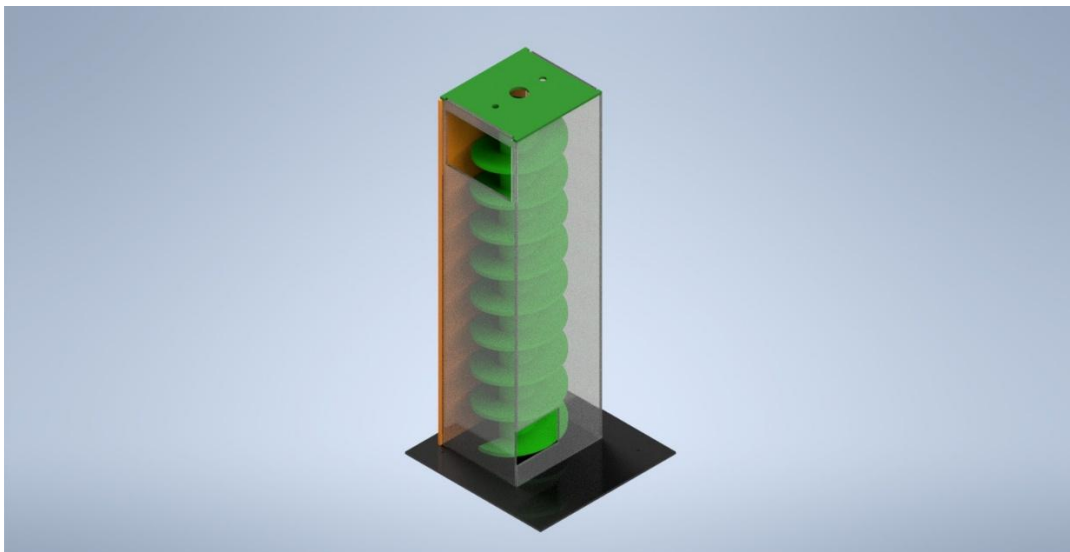


Figure 1 – Archimedes Screw Lifting Mechanism (Autodesk Inventor)

Shooting Mechanism – Rotating Arm

A 54 mm rotating arm (mass 3 g) is driven by a DC motor through a 180° swing in 0.2 s. The arm strikes the ball (mass 7 g, diameter 38 mm) at its tip, imparting a launch velocity of 1.13 m/s at 20° elevation. Projectile calculations give a range of 84 mm and a maximum height of 7.6 mm. Required $T_m = 2.01 \times 10^{-3}$ Nm; $T_{RMS} = 1.17 \times 10^{-3}$ Nm.

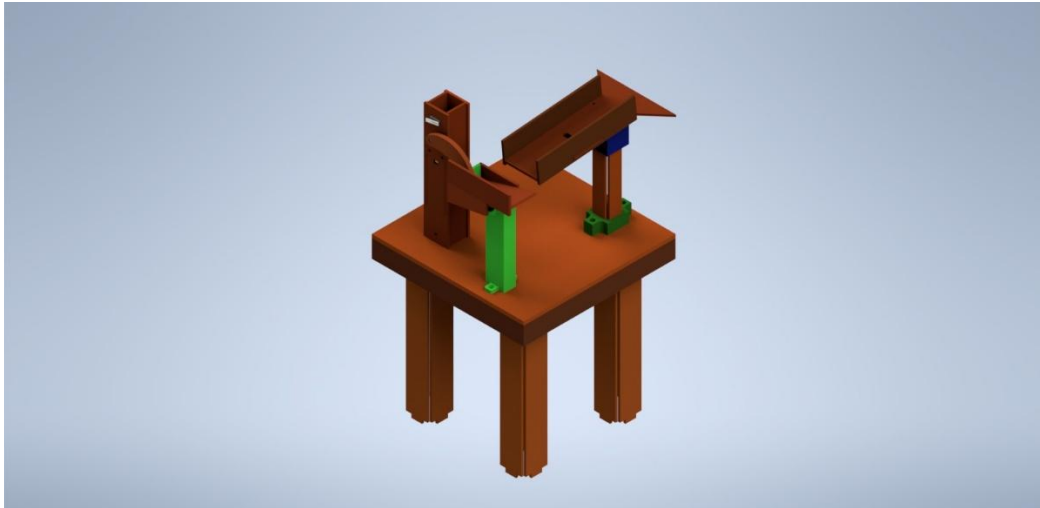


Figure 2 – Shooting Arm Mechanism (Autodesk Inventor)

Sorting / Guiding Mechanism – Movable Slider

A rack-and-pinion system (rack mass 4 g, travel 130 mm, pinion pitch radius 6 mm, 12 teeth) moves the decision slider horizontally in 2 s. A stepper motor provides precise positional control driven by the joystick input. $T_{RMS} = 0.0012$ Nm, well within the stepper's rated output. The slider directs each incoming ball to either a 'score' slot or a 'miss' slot based on the player's reaction to the ball's colour.

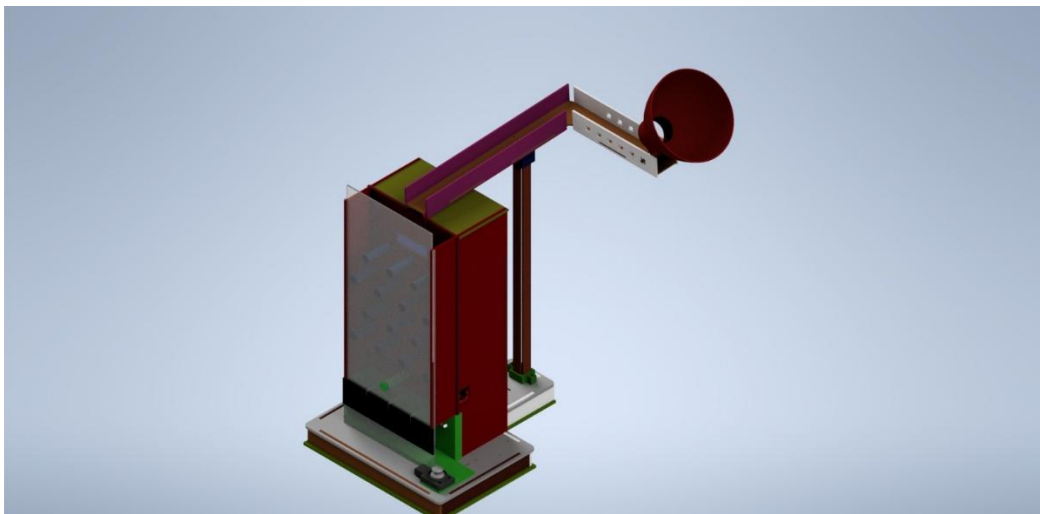


Figure 3 – Rack-and-Pinion Sorting Mechanism (Autodesk Inventor)

Full CAD Assembly

The complete 3D assembly was modelled in Autodesk Inventor, with overall dimensions of 94.5 cm (W) × 70 cm (L) × 72 cm (H). All mechanisms were verified for clearance and assembly fitment within the toy enclosure.

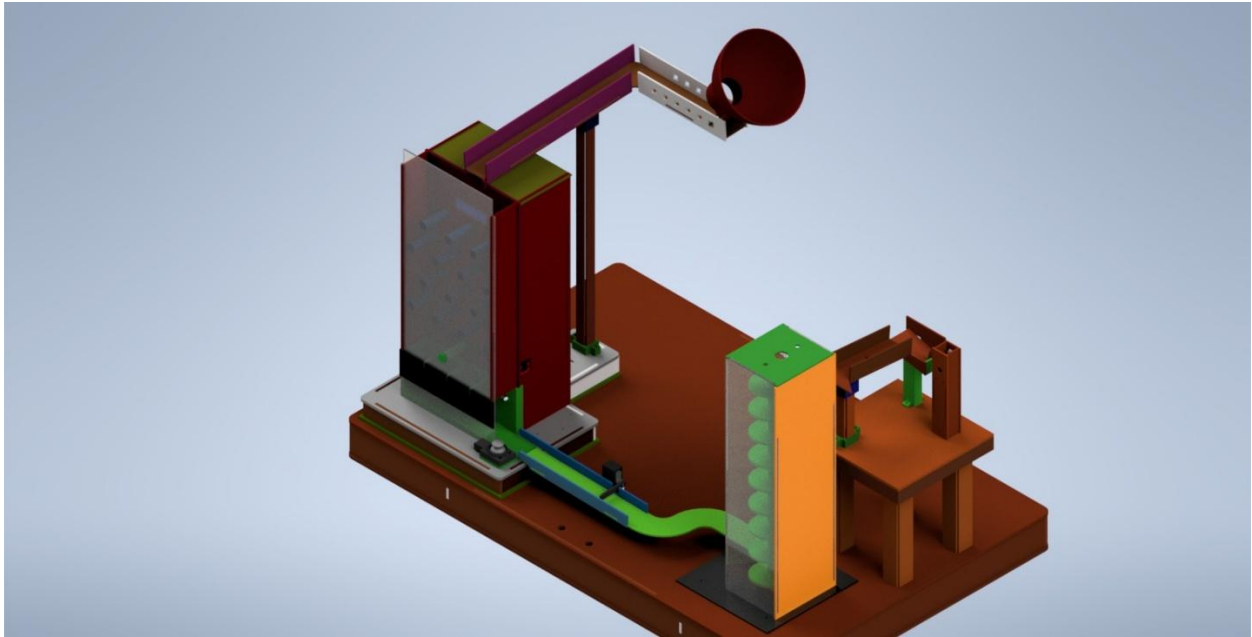


Figure 4 – Full System CAD Assembly (Autodesk Inventor)

5.2 Electrical and Control Elements

The electrical system is built around two Arduino microcontrollers communicating via a serial link. One Arduino manages the sensor inputs and shooting/lifting control; the other handles the sorting logic, display, and Bluetooth communication.

Sensor System

IR sensors (operating at 3.3–5 V, 20 mA) are placed at three critical points: (1) at the screw entry to trigger lifting, (2) at the shooter entry to confirm ball readiness, and (3) at the colour detection zone before the slider. The colour sensor distinguishes white from black balls by comparing reflected IR intensity against a threshold.

Motor Control & Wiring

Each DC motor is driven via an H-bridge driver.

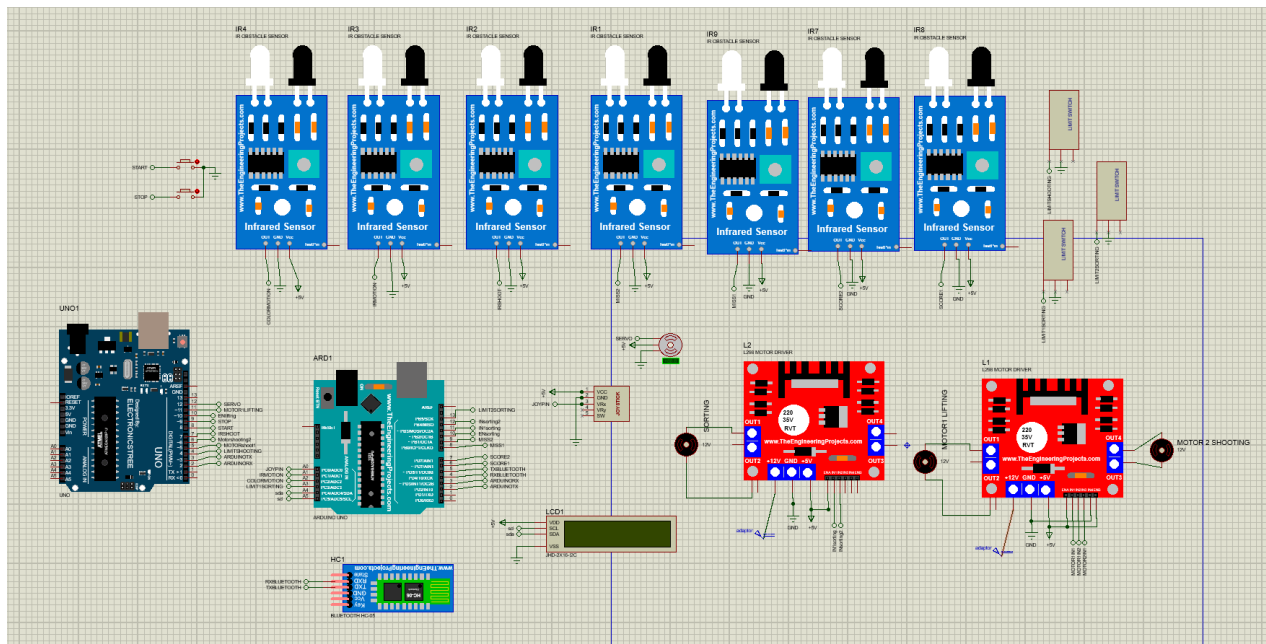


Figure 5 – Full Wiring Diagram (Proteus)

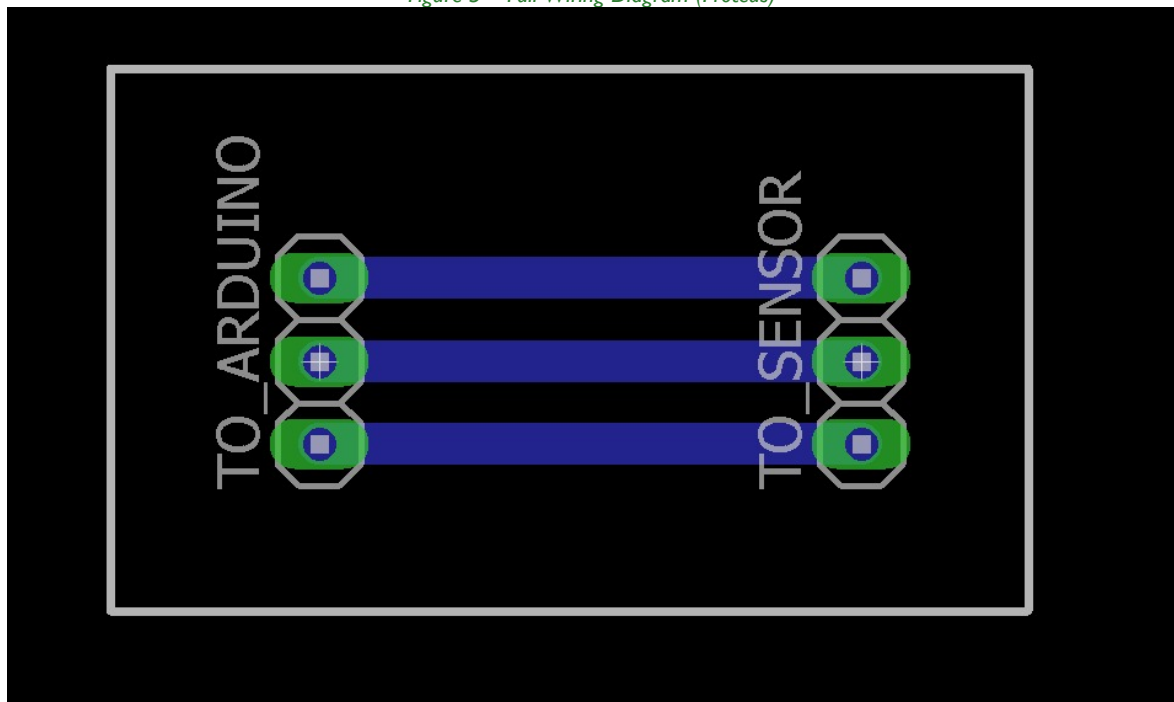


Figure 6 – Custom PCB Layout (Proteus)

Block Diagram

The system block diagram illustrates the signal flow from sensors through the Arduinos to each actuator and the display. Bluetooth communication allows an optional mobile app to display real-time scores.

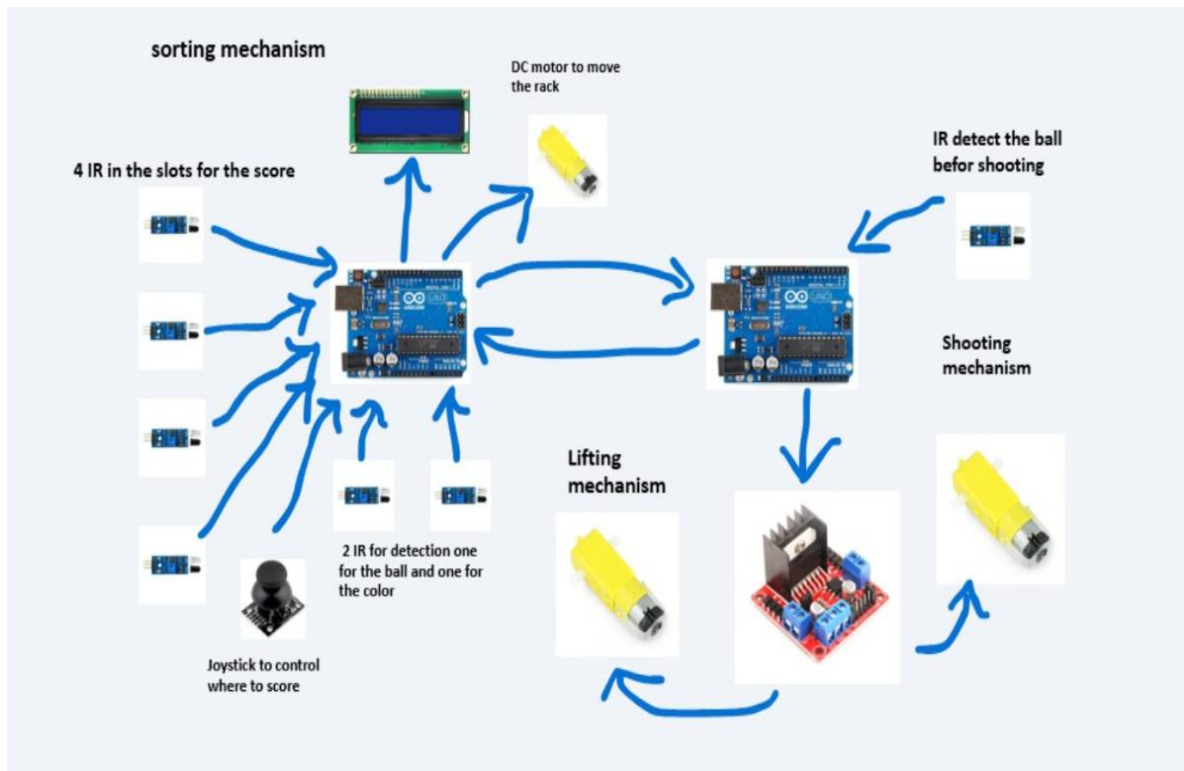


Figure 7 – System Block Diagram

6. Safety Considerations

The toy has been designed in compliance with general child product safety guidelines applicable to toys for children aged 2–10. The following specific design decisions ensure a safe user experience:

Mechanical Safety

- All external surfaces are rounded with fillets ≥ 3 mm; no sharp edges or protruding fasteners are accessible to a child.
- Ball diameter is 30 mm — well above the 3D choke-hazard threshold of 31.7 mm for children under 3 (borderline; a slightly larger ball is recommended for the under-3 segment).
- Maximum ball launch velocity is 1.13 m/s — insufficient to cause injury on skin contact at the tested range.
- The Archimedes screw is fully enclosed within a transparent polycarbonate tube to prevent finger insertion during operation.

Electrical Safety

- All wiring is routed internally and secured with cable ties; no bare conductors are accessible from outside the housing.
- All electrical components are housed.
- The operating voltage is 12 V DC supplied by a wall adapter; no mains-level voltages are present inside the toy.
- PCB safety circuits prevent motor overload and short-circuit conditions.

Material Safety

- PLA plastic (used for 3D-printed parts) is biodegradable and non-toxic.
- Laser-cut plywood edges are sanded and sealed to prevent splinters.

7. Educational Impact

The toy is designed to develop a range of cognitive and physical skills appropriate for children aged 2–10, while keeping the experience fun and self-motivating through the score-based reward system.

Cognitive Development

- **Colour Recognition:** children must rapidly identify ball colour (white or black) and make a decision — strengthening colour discrimination and pattern recognition.
- **Cause and Effect:** the direct link between slider position and ball routing builds logical reasoning; children observe that their choice has a measurable, scored consequence.
- **Numerical Literacy:** the LCD score display introduces basic counting and positive/negative number concepts in a game context.
- **Attention and Concentration:** the time-limited nature of each ball cycle demands sustained focus and improves attention span over repeated play sessions.

Motor Skills

- **Fine Motor Control:** joystick manipulation to position the slider precisely develops hand-eye coordination and fine motor dexterity.
- **Reaction Time:** the shooting interval trains reflex responses; with practice, children improve their timing and accuracy.

Introduction to STEM Concepts

- **Physics:** visible ball trajectory illustrates projectile motion, gravity, and energy transfer in an intuitive, hands-on way.
- **Electronics Awareness:** the presence of sensors, displays, and motors in a transparent or partially-visible enclosure demystifies technology for young users.
- **System Thinking:** the three-stage sequence (lift → shoot → sort) exposes children to the concept of interconnected subsystems working together.

Overall, the toy bridges play and education, encouraging children to think, react, and improve with every session — aligning with modern STEM education goals.

8. Work Breakdown Structure (WBS)

The following WBS outlines all major tasks from concept to final documentation. Each level represents a decomposition of project scope into manageable work packages.

Level 1 – Project Overview

- I.1 Conceptual Design
 - I.1.1 Define user needs and project scope
 - I.1.2 Research similar projects for inspiration
 - I.1.3 Sketch the overall design

Level 2 – Mechanical Design

- I.2.1 Lifting Mechanism Design
- I.2.2 Shooting Mechanism Design
- I.2.3 Sorting Mechanism Design
- I.2.4 System Assembly Design

Level 3 – Electrical Design

- I.3.1 Sensor Integration
- I.3.2 Display System
- I.3.3 Custom PCB Design
- I.3.4 Communication Systems (Bluetooth)
- I.3.5 Wiring Diagram Creation
- I.3.6 Arduino Communication Setup

Level 4 – Software Development

- I.4.1 Ball Lifting Control
- I.4.2 Shooting Control
- I.4.3 Sorting Control
- I.4.4 Bluetooth Communication
- I.4.5 Display Interface
- I.4.6 Controllable Column
- I.4.7 Arduino Communication Code

Level 5 – Prototyping & Integration

- I.5.1 Mechanical Prototyping
- I.5.2 Electrical Prototyping
- I.5.3 System Integration

Level 6 – Testing & Optimization

- I.6.1 Prototype Testing
- I.6.2 Scoring System Testing

Level 7 – Final Documentation

- I.7.1 Product Design Specification (PDS)
- I.7.2 Assembly Instructions
- I.7.3 Software Code Documentation
- I.7.4 Final Project Report

9. Product Design Specifications (PDS)

Specification Category	Details
Product Overview	Interactive mechatronic toy for children aged 2–10. Integrates shooting, sorting, and lifting mechanisms.
Performance	Three key functions: (1) projectile shooting, (2) Archimedes screw lifting, (3) colour-based sorting. Includes score/miss slots and live score display.
Cost	~14,000 L.E. (3D-printed parts + laser-cut wood + electronic components)
Customers	Children aged 2–10 and their parents/educators.
Safety	No sharp edges, no small detachable parts. All electronics enclosed in child-safe housing, wiring hidden internally.
Mechanical Functionality	DC motor shoots the ball. Archimedes screw lifts ball (DC motor). Sorting via IR sensor + stepper-driven slider.
Electrical Integration	Sensors detect ball colour and position; signals processed by Arduino; motors controlled via H-bridge and stepper driver.
Software Control	Arduino manages sensor input, motor actuation, sorting logic, display, and Bluetooth communication with mobile app.
Manufacture	3D-printed PLA parts + laser-cut plywood. Electronic components purchased off-the-shelf.
Size	94.5 cm (W) × 70 cm (L) × 72 cm (H). Weight is suitable for adult-assisted child use.
Aesthetic	Colourful, rounded, child-friendly design. Dark navy and grass-green colour scheme with transparent acrylic panels.
Maintenance	Easily replaceable parts; clean internal wiring layout for serviceability.
Working Voltage / Current	IR sensor: 3.3–5 V, 20 mA DC motor: 4–9 V, 190 mA (no load)

10. Actuator Sizing

This section presents the complete actuator sizing calculations for all three subsystems: the Lifting Mechanism (Archimedes screw), the Shooting Mechanism (rotating arm), and the Sorting Mechanism (rack-and-pinion slider). A trapezoidal motion profile is used throughout.

10.1 Lifting Mechanism – Actuator Sizing

General Equation

$$J_{\text{total}} * \theta_{\text{double_dot}} = T_m - T_r$$

$$\text{Therefore: } T_m = J_{\text{total}} * \theta_{\text{double_dot}} + T_r$$

Parameters

- Screw mass = 0.3 kg
- Outer diameter = 100 mm → radius = 0.05 m
- Length = 0.4 m
- Pitch = 0.045 m/rev
- Ball mass = 0.007 kg

Inertia of the Screw

$$J = (1/2) * m * r^2$$

$$J = (1/2) * (0.30) * (0.05)^2$$

$$J = 0.000375 \text{ kg m}^2 = 3.75 \times 10^{-4} \text{ kg m}^2$$

Target Speed & Acceleration

Target speed is 100 rpm; the yellow motor reaches this speed in 0.5 s.

$$\omega = (2 * \pi * 100) / 60 = 10.47 \text{ rad/s}$$

$$\alpha = \omega / t_{\text{accel}} = 10.47 / 0.5 = 20.94 \text{ rad/s}^2$$

Torque Required at the Screw

The force on the ball and the resistance torque (using screw efficiency $\eta = 30\%$):

$$F_{\text{ball}} = m * g = 0.007 * 9.81 = 0.0687 \text{ N}$$

$$T_r = F / (2 * \pi * \eta * P) = 0.0687 / (2 * \pi * 0.3 * 0.045)$$

$$T_r = 1.6 \times 10^{-3} \text{ Nm} \approx 0.02 \text{ Nm}$$

Motor Selection Check

$$T_m = J_{\text{total}} * \alpha + T_r$$

$$T_m = (3.75 \times 10^{-4}) * (20.94) + 0.02$$

$$T_m = 0.0278 \text{ Nm} \approx 0.03 \text{ Nm}$$

The yellow motor provides 0.08 – 0.15 Nm stall torque. Required $T_m = 0.03 \text{ Nm}$ → Motor is SUITABLE.

10.2 T_RMS Calculations – Lifting Mechanism

Parameters

- Lift distance = 0.4 m, Total time = 7 s
- Pitch = 0.045 m/rev, $J_{\text{screw}} = 3.75 \times 10^{-4} \text{ kg m}^2$, $T_r = 0.02 \text{ Nm}$

Total Revolutions & Angular Distance

$$N = 0.4 / 0.045 = 8.89 \text{ rev}$$

$$\theta_{\text{total}} = 8.89 * (2 * \pi) = 55.85 \text{ rad}$$

Trapezoidal Motion Profile

- Acceleration phase: $t_a = 1.75 \text{ s}$ (25% of profile)
- Constant-speed phase: $t_c = 3.5 \text{ s}$ (50% of profile)
- Deceleration phase: $t_d = 1.75 \text{ s}$ (25% of profile)

Maximum Angular Speed

$$\begin{aligned}\theta &= \omega_{\text{max}} * (t_c + (t_a + t_d) / 2) \\ 55.85 &= \omega_{\text{max}} * (3.5 + 1.75) \\ \omega_{\text{max}} &= 55.85 / 5.25 = 10.64 \text{ rad/s}\end{aligned}$$

Angular Acceleration

$$\alpha = \omega_{\text{max}} / t_a = 10.64 / 1.75 = 6.08 \text{ rad/s}^2$$

Torque Analysis

$$\begin{aligned}T_a &= J * \alpha + T_r = (3.75 \times 10^{-4})(6.08) + 0.02 = 0.0223 \text{ Nm} \\ T_c &= T_r = 0.0200 \text{ Nm} \\ T_d &= T_r - J * \alpha = 0.02 - (3.75 \times 10^{-4})(6.08) = 0.0177 \text{ Nm}\end{aligned}$$

Final T_RMS

$$\begin{aligned}T_{\text{RMS}} &= \sqrt{(T_a^2 * t_a + T_c^2 * t_c + T_d^2 * t_d) / T_{\text{total}}} \\ T_{\text{RMS}} &= \sqrt{(0.0223^2 * 1.75 + 0.02^2 * 3.5 + 0.0177^2 * 1.75) / 7} \\ T_{\text{RMS}} &= 0.0202 \text{ Nm}\end{aligned}$$

10.3 Shooting Mechanism – Actuator Sizing

Parameters

- Arm: Length = 54 mm = 0.054 m, Mass = 3 g = 0.003 kg
- Ball: Diameter = 3 cm, Mass = 7 g = 0.007 kg
- Motion: Rotation angle = $180^\circ = \pi \text{ rad}$, Total move time = 0.2 s
- Profile times: $t_a = 0.05 \text{ s}$, $t_c = 0.10 \text{ s}$, $t_d = 0.05 \text{ s}$

Maximum Angular Speed

$$\begin{aligned}\theta &= \omega_{\text{max}} * (t_c + (t_a + t_d) / 2) \\ \pi &= \omega_{\text{max}} * (0.10 + 0.05) \\ \omega_{\text{max}} &= \pi / 0.15 = 20.94 \text{ rad/s}\end{aligned}$$

Angular Acceleration

$$\alpha = \omega_{\text{max}} / t_a = 20.94 / 0.05 = 418.9 \text{ rad/s}^2$$

Arm Inertia & Resistance Torque

$$\begin{aligned}J_{\text{arm}} &= (1/3) * m * L^2 = (1/3)(0.003)(0.054)^2 = 2.92 \times 10^{-6} \text{ kg} \cdot \text{m}^2 \\ T_{r_{\text{arm}}} &= m * g * r = (0.003)(9.81)(0.027) = 7.9 \times 10^{-4} \text{ Nm}\end{aligned}$$

Motor Torque Required

$$\begin{aligned}T_m &= J_{\text{arm}} * \alpha + T_{r_{\text{arm}}} \\ T_m &= (2.92 \times 10^{-6})(418.9) + 7.9 \times 10^{-4} \\ T_m &= 2.01 \times 10^{-3} \text{ Nm}\end{aligned}$$

T_RMS Calculations – Shooting

$$\begin{aligned}T_a &= J * \alpha + T_r = 2.01 \times 10^{-3} \text{ Nm} \\ T_c &= T_r = 7.9 \times 10^{-4} \text{ Nm} \\ T_d &= T_r - J * \alpha = 7.9 \times 10^{-4} - (2.92 \times 10^{-6} * 418.9) = -4.33 \times 10^{-4} \text{ Nm}\end{aligned}$$

$$T_{\text{RMS}} = \sqrt{(T_a^2 \times 0.05 + T_c^2 \times 0.10 + T_d^2 \times 0.05) / 0.2}$$
$$T_{\text{RMS}} = 1.17 \times 10^{-3} \text{ Nm}$$

10.4 Projectile Calculations

Launch Conditions

- Launch velocity: $v = \omega_{\text{max}} \times r = 20.94 \times 0.054 = 1.13 \text{ m/s}$
- Launch angle: $\theta = 20^\circ$
- Gravitational acceleration: $g = 9.81 \text{ m/s}^2$

Equations of Motion

$$x(t) = v \cdot \cos(\theta) \cdot t$$
$$y(t) = v \cdot \sin(\theta) \cdot t - (1/2) \cdot g \cdot t^2$$

Time of Flight

$$t_{\text{flight}} = (2 \cdot v \cdot \sin(\theta)) / g$$
$$t_{\text{flight}} = (2 \cdot 1.13 \cdot \sin(20^\circ)) / 9.81 = 0.079 \text{ s}$$

Horizontal Range

$$x = v \cdot \cos(\theta) \cdot t = 1.13 \cdot \cos(20^\circ) \cdot 0.079 = 0.084 \text{ m} = 84 \text{ mm}$$

Maximum Height

$$H_{\text{max}} = (v^2 \cdot \sin^2(\theta)) / (2 \cdot g)$$
$$H_{\text{max}} = (1.13^2 \cdot \sin^2(20^\circ)) / (2 \cdot 9.81) = 0.0076 \text{ m} = 7.6 \text{ mm}$$

10.5 Sorting Mechanism – Actuator Sizing

Parameters

- Rack load: mass = 0.004 kg, travel distance = 0.13 m, total time = 2 s
- Pinion: 12 teeth, pitch radius $r = 0.006 \text{ m}$
- Profile: $t_a = 0.5 \text{ s}$, $t_c = 1.0 \text{ s}$, $t_d = 0.5 \text{ s}$

Linear Speed & Acceleration

$$x = v_{\text{max}} \cdot (t_c + (t_a + t_d) / 2)$$
$$0.13 = v_{\text{max}} \cdot (1.0 + 0.5) \Rightarrow v_{\text{max}} = 0.0867 \text{ m/s}$$
$$a = v_{\text{max}} / t_a = 0.0867 / 0.5 = 0.173 \text{ m/s}^2$$

Angular Motion at Pinion

$$\omega = v / r_{\text{pinion}} = 0.0867 / 0.006 = 14.45 \text{ rad/s}$$
$$\alpha = a / r = 0.173 / 0.006 = 28.83 \text{ rad/s}^2$$

Inertia Calculations

$$J_{\text{load}} = m \cdot r^2 = (0.004)(0.006)^2 = 1.44 \times 10^{-7} \text{ kg} \cdot \text{m}^2$$
$$J_{\text{pinion}} = 2.00 \times 10^{-7} \text{ kg} \cdot \text{m}^2 \text{ (manufacturer)}$$
$$J_{\text{total}} = 1.44 \times 10^{-7} + 2.00 \times 10^{-7} = 3.44 \times 10^{-7} \text{ kg} \cdot \text{m}^2$$

Torque Analysis

Note: Additional friction torque $T_r = 0.0012 \text{ Nm}$ due to 3D-printed surfaces ($F_{\text{friction}} = 0.2 \text{ N}$ assumed).

$$T_a = J_{\text{total}} \cdot \alpha + T_r = (3.44 \times 10^{-7})(28.83) + 0.0012 = 0.00121 \text{ Nm}$$
$$T_c = T_r = 0.00120 \text{ Nm}$$
$$T_d = T_r - J_{\text{total}} \cdot \alpha = 0.00119 \text{ Nm}$$

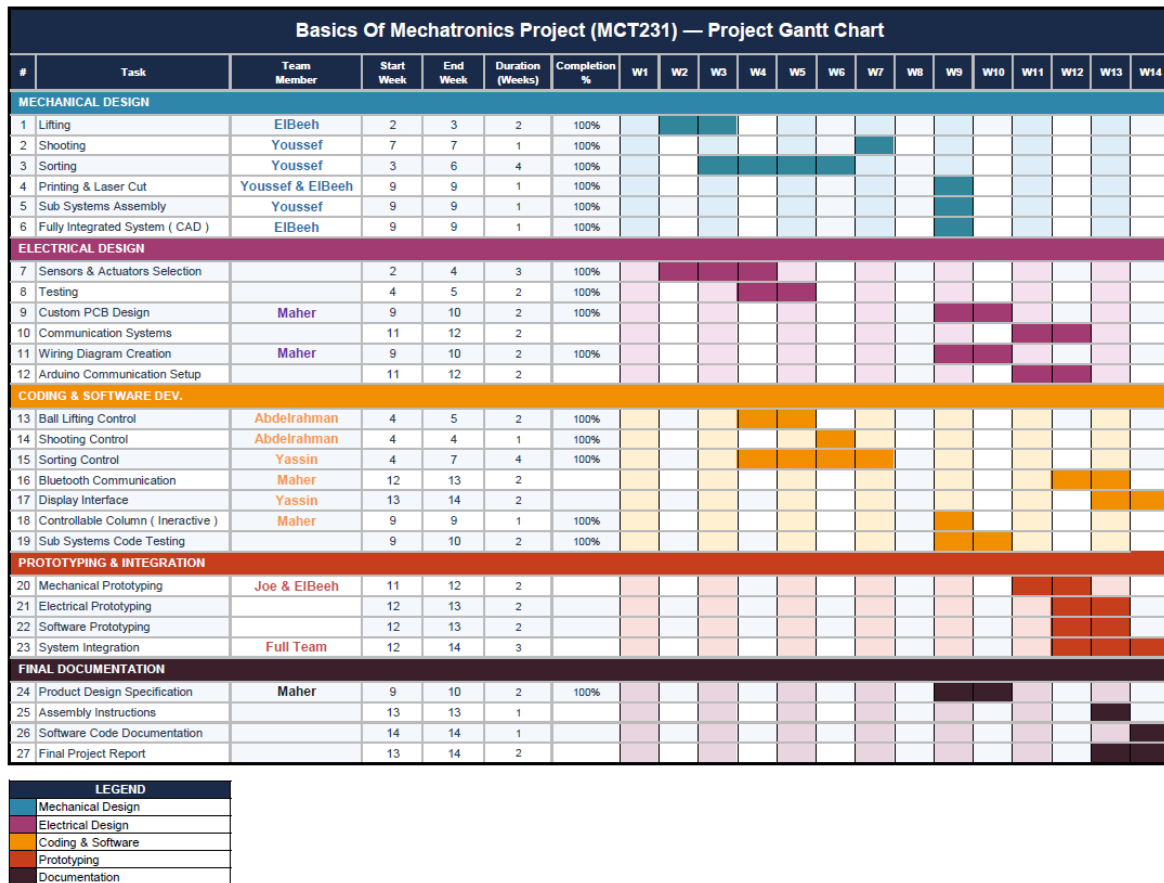


Figure 9 – Project Gantt Chart

I2. Bill of Materials

The following tables list all components required for each subsystem. Quantities and specifications are based on the final design.

I2.1 Lifting Subsystem

I2.2 Shooting Subsystem

ITEM	QTY	PART NUM	DESCRIPTION
1		1	Base
2		1	Inclined Ramp
3		1	Motor Fixation .1
4		2	Motor Fixation .2
5		2	Ramp Side Wall
6		1	Path
7		2	Path Side Wall
8		1	Shooting Arm
9		1	Lower Path Fixation
10		1	Upper Path Fixation
11		4	Path Holder
12		1	IR Sensor
13		1	Motor Fixation .1 FV
14		1	DC Gear Motor Single Shaft
15		1	wall extension
16		1	ramp extension
17		1	Limit_Switch
18		1	Box Of Fixation
19		16	Assembly Fixation Rod

I2.3 Sorting / Guiding Subsystem

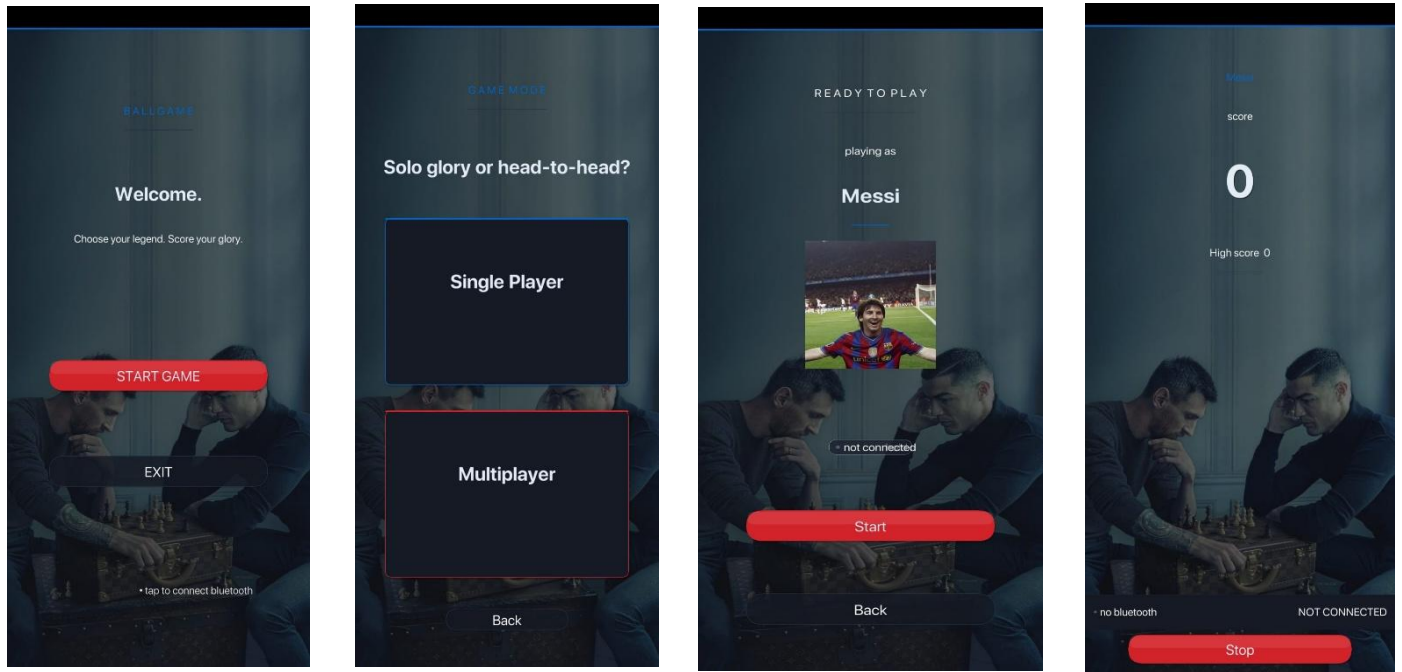
ITEM	QTY	PART NUN DESCRIPTION
1		1 Base
2		1 Random Plate (Laser Cut)
3		14 Randomization Stick
4		2 Finishing Sheets .1 (Right)
5		1 Rack (Female Fixation)
6		1 Rack (Male Fixation
7		1 Sorting Mechanism Base Fixation
8		1 Finishing Sheets .2 (Upper)
9		1 Finishing Sheets .3 (Back)
10		1 Finishing Sheets .4 (Front) Acrylic
11		1 Finishing Sheets .5 (Buttom)
12		1 Joystick Shield
13		2 Base Leg V1
14		1 Base Leg V2
15		1 Movable Stick
16		1 Lower Path Fixation
17		4 Path Holder
18		1 Upper Path Fixation
19		1 Path
20		1 Path Side Wall
21		1 Path Back End
22		1 Cone FV
23		2 Path Base Fixation .1
24		1 Path Base Fixation .2
25		1 Sensors Shield
26		1 DC Gear Motor Single Shaft
27		6 IR Sensor
28		2 Limit_Switch
29		1 Joystick
30		1 Base Leg V2 Slotted
31		1 DC Motor Fixing Wall
32		1 Geared Motor Transmission
33		1 Pinion Gear FV
34		1 Path Plate Connector
35		1 LCD-I2C_1 STEP AP242
36		1 Extension .1
37		1 Extension.2
38		1 Extension .3

12.4 Electrical & Control Subsystem

Item	Description	Qty	Spec / Notes
Arduino Uno (×2)	Main microcontrollers	2	Serial communication between them
Custom PCB	Designed in Proteus; integrates drivers	1	H-bridge, LCD driver, Bluetooth
HC-05 Bluetooth Module	Wireless communication to mobile app	1	UART, 3.3 V logic
LCD Display (16×2)	Score display	1	I2C interface, 5 V
IR Sensors	Ball detection at multiple positions	7	3.3–5 V, 20 mA each
L298N H-Bridge (×2)	DC motor driver modules	2	One per 2 DC motors
Power Supply / Adapter	12 V DC wall adapter	1	Min 2 A, centre-positive
Joy stick	Move controllable stick	1	-----
Limit switch	Restricts motor travel	3	-----
button	Start and end buttons	2	-----

13. Mobile Application Screenshots

The companion mobile application (Android/iOS) connects to the toy via Bluetooth (HC-05) and provides real-time score display, session history, and system status. Representative screenshots are presented below.



I4. GitHub Repository – QR Code

All source code, CAD files, PCB designs, and documentation are hosted on the team GitHub repository. Scan the QR code below to access the repository directly.



The repository is organised into the following folders:

- **/Software**
- **/Mechanical**
- **/Electrical**
- **/App**
- **/Docs**

15. Conclusion

This report has documented the full design, analysis, and implementation of an interactive mechatronic toy — a colour-sorting ball game — developed as part of MCT231: Basics of Mechatronics at the Faculty of Engineering, Ain Shams University.

The system successfully integrates three interdependent mechanical subsystems (Archimedes screw lifting, rotating-arm shooting, and rack-and-pinion sorting) with a dual-Arduino control architecture, custom PCB, Bluetooth connectivity, and a companion mobile application. All actuator sizing calculations confirmed that the selected DC yellow motor are well within their rated torque limits, with comfortable safety margins across all operating phases.

Key engineering outcomes achieved:

- Lifting mechanism: Required $T_m = 0.03 \text{ Nm}$ (motor rated at 0.08–0.15 Nm); $T_{\text{RMS}} = 0.0202 \text{ Nm}$ over a 7-second lift cycle.
- Shooting mechanism: Launch velocity of 1.13 m/s at 20° elevation yields a range of 84 mm; $T_{\text{RMS}} = 1.17 \times 10^{-3} \text{ Nm}$.
- Sorting mechanism: Rack-and-pinion drive achieves precise slider positioning; $T_{\text{RMS}} = 0.0012 \text{ Nm}$.
- All external surfaces are child-safe, all electronics are fully enclosed, and operating voltage remains at 12 V DC.

The product meets all specifications defined in the PDS (Section 9), including the target footprint, cost estimate (~14,000 L.E.), age-group safety requirements, and educational objectives across cognitive development, motor skill training, and STEM concept introduction.

Overall, the project demonstrated the successful application of mechatronics principles — combining mechanical design, electronics, control software, and human interaction — to produce a functional, safe, and educationally valuable prototype.